DARKHAN MENTOR

Research Methodology Explained: Qualitative, Quantitative and Mixed Methods

A Thesis Guide

How to choose a research approach, collect sound evidence, analyse it honestly, and write a methodology chapter that examiners can trust.

Darkhan Mentor

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This guide is written for undergraduate and postgraduate students who are planning a thesis, dissertation or final-year project. It explains ideas in plain words, with worked examples and checklists you can use at once.

Abstract

Every thesis stands or falls on its methodology. A clear research question, a suitable design, careful data collection and honest analysis are what turn opinions into evidence. Yet methodology is the chapter many students find hardest to write, because it asks them to justify choices they have often made by habit, by copying a previous thesis, or by following what a supervisor suggested without fully understanding why.

Later chapters deal with the decisions that cut across all approaches: sampling and sample size, validity, reliability and trustworthiness, and research ethics. A decision guide helps the reader match a research question to an approach, and a final chapter collects the most common mistakes — from mismatched questions and methods to over-claiming from small samples — together with a checklist to use before submission. The aim throughout is practical: after reading this guide, a student should be able to defend every methodological choice in a viva, and a reader should be able to see exactly how the findings were produced.

Keywords: research methodology; qualitative research; quantitative research; mixed methods; thematic analysis; sampling; validity; research ethics; thesis writing.

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How to use this guide

Read Chapters 1 and 2 first, then go to the chapter that matches your approach — Chapter 3 for qualitative, Chapter 5 for quantitative, Chapter 7 for mixed methods. Use Chapter 11 as a decision check before you finalise your design, and Chapter 12 as a last review before submission.

Chapter 1: Introduction — What Methodology Means and Why It Matters

Methodology is the reasoned account of how a study was done and why it was done that way. It is not a list of tools. A list says: I used a questionnaire. A methodology says: I used a questionnaire because my question asked how widespread a pattern was, because a survey could reach many people quickly, because a standardised instrument would let me compare groups, and because the alternatives — interviews with a handful of people, or an experiment I could not control — would not have answered that question. The first version describes an activity. The second defends a decision. Examiners look for the second.

Three words are often confused, and keeping them apart makes the whole subject clearer. A method is a single technique: an interview, a survey, a statistical test. Methodology is the logic that connects the research question to the choice of methods, the way data are collected and analysed, and the claims that can finally be made. A research design is the overall plan — the structure that holds the methods together, such as a case study, an experiment, or a survey design. A thesis that confuses these terms usually ends up describing tools without explaining why they were chosen, and that is the most common weakness in a methodology chapter.

Why methodology matters

First, methodology is what makes findings believable. Two students can study the same topic and reach opposite conclusions, and the difference is usually not honesty but method: one asked leading questions, the other did not; one sampled only willing volunteers, the other sampled systematically; one tested a claim against a comparison group, the other had no comparison at all. When the method is reported clearly, a reader can judge for themselves whether the conclusion follows from the evidence.

A worked example: one topic, three questions

Consider the topic of student dropout from evening classes at a college. The topic alone does not decide the method. The research question does. If the question is How many evening students leave before the end of term, and which groups leave most often?, the study needs numbers from college records and a survey — a quantitative approach. If the question is Why do evening students decide to leave, in their own words?, the study needs in-depth interviews — a qualitative approach. If the question is How many leave, and what reasons explain the pattern?, the study needs both numbers and interviews — a mixed methods approach. The same topic, three different questions, three different methodologies. That is why a thesis should always begin its methodology chapter by restating the research question, and then showing how each choice follows from it.

The golden rule

Every methodological choice in your thesis should be traceable to your research question. If you cannot explain, in one or two sentences, why a method helps answer your question better than the alternatives, you have not yet justified it — you have only described it.

Chapter 2: Research Paradigms: Positivism, Interpretivism and Pragmatism

A paradigm is a set of basic beliefs about what the world is like and how we can know anything about it. Researchers do not always name their paradigm, but they always have one, and it quietly shapes every decision they make: what counts as evidence, what a good explanation looks like, and what the researcher is allowed to claim. Three paradigms dominate thesis research in the social sciences, education, health and business: positivism, interpretivism and pragmatism.

Positivism

Positivism assumes that there is a single, objective reality that exists independently of what people think about it, and that this reality can be measured. A positivist researcher tries to stand apart from what is studied, to measure variables precisely, to control other influences, and to test hypotheses with numbers. The ideal is the natural-science model: observe, measure, test, generalise. A study that asks Does class size affect examination scores? and tests the relationship with data from a hundred schools is working in a positivist spirit, whether or not the researcher uses the word.

Interpretivism

Interpretivism assumes that social reality is made by people through meaning: the same event can mean different things to different people, and those meanings shape what happens next. An interpretivist researcher therefore tries to understand the world from the participants' point of view, usually through interviews, observation and close reading of what people say and do. The goal is depth and understanding rather than measurement and generalisation. A study that asks How do first-generation university students experience their first year? is interpretivist in spirit.

Pragmatism

Pragmatism refuses to choose between the two. Its central idea is that the research question should decide the method, not a philosophical commitment made in advance. If numbers answer the question best, use numbers; if interviews answer it best, use interviews; if both are needed, use both. Pragmatists judge a method by

whether it works for the problem at hand. This is the paradigm most often cited by mixed methods researchers, and it is the most common position in applied thesis research, even when students do not name it.

Pragmatism is practical, but it is not an excuse for carelessness. A pragmatist still has to justify each choice, still has to collect and analyse data rigorously, and still has to be honest about what each kind of evidence can and cannot show. Choosing methods because they are convenient, or because they are familiar, is not pragmatism; it is habit, and examiners will notice the difference.

Why paradigms matter for your thesis

You do not need a long philosophical discussion in your methodology chapter. A short, honest paragraph is enough: state the paradigm that fits your study, explain in two or three sentences what it assumes, and show how it connects to your question and your methods. For example: This study adopts an interpretivist stance, because its aim is to understand how teachers interpret a new curriculum in their own classrooms, and that aim requires close attention to meaning rather than measurement. Or: This study adopts a pragmatic stance, because it asks both how widespread a problem is and why it occurs, and answering both parts requires numbers and interviews together.

Common mistake

Do not claim a paradigm you do not follow. A thesis that declares itself positivist and then reports only five unstructured interviews, with no measurement and no hypothesis, has a paradigm and a method that contradict each other. Examiners notice that contradiction at once. It is far better to name the paradigm that actually matches your design — even if it is simply pragmatism — than to borrow an impressive-sounding label.

Chapter 3: Qualitative Research: Interviews, Focus Groups, Observation and Case Studies

Qualitative research studies people in their own terms. It asks how they understand their lives, what their experiences mean to them, and how social processes unfold in real settings. Its data are usually words, images and observations rather than numbers, and its analysis looks for patterns of meaning rather than statistical relationships. It is the right choice when a question asks how or why, when little is known about a topic, or when the aim is depth rather than breadth.

Interviews

The interview is the most common qualitative method, and the most flexible. In a structured interview, every participant is asked the same questions in the same

order, which makes comparison easy but limits depth. In a semi-structured interview — the usual choice for a thesis — the researcher has a guide with key questions and prompts, but can follow up interesting answers, ask for examples, and change the order when the conversation requires it. In an unstructured interview, the researcher begins with a broad topic and lets the participant lead. The more structure, the easier the comparison; the less structure, the richer the story. Most theses use semi-structured interviews, because they balance the two.

Focus groups

A focus group brings six to ten people together to discuss a topic, guided by a moderator. Its value is interaction: one person's comment prompts another's memory, disagreement reveals different views, and shared laughter or silence shows what a group takes for granted. Focus groups are excellent for exploring attitudes towards a new service, a curriculum change or a community issue, and for generating ideas quickly. They are less suitable when a topic is sensitive or private, because people may not speak openly in front of others, and because one loud voice can dominate. The moderator's job is to keep the discussion on track, draw out quieter participants, and prevent the group from becoming an interview with an audience.

Observation

Observation means watching what people actually do, rather than asking them what they do. In participant observation, the researcher joins the setting — sitting in a classroom, working alongside staff — and records what happens. In non-participant observation, the researcher watches from the edge, taking structured notes. Observation is powerful because what people say and what they do often differ: a teacher may say they use group work every lesson, while observation shows it happens once a month. The challenge is that the researcher's presence changes behaviour, at least at first, and that observation is time-consuming. Field notes should separate description (what happened) from interpretation (what the researcher thinks it meant), and both should be written the same day, while memory is fresh.

Case studies

Worked example — choosing a qualitative method

Question: How do new nurses cope with night shifts in their first year? Interviews are the best fit, because coping is personal and partly private; a focus group might silence honest talk about fear or mistakes. Add two weeks of observation on the ward, and the study can compare what nurses say they do with what they actually do. That combination — interviews plus observation — is stronger than either method alone.

Chapter 4: Analysing Qualitative Data: Thematic Analysis Step by Step

Thematic analysis is the most widely used way of analysing qualitative data in a thesis. It is a method for identifying, analysing and reporting patterns — themes — within data. It is flexible, it works with interviews, focus groups and open-ended survey answers, and it can be learned and applied rigorously by a student researcher. The steps below follow the approach described in the standard literature on thematic analysis, and they are illustrated with a small worked example so the reader can see each move.

Step 1 — Familiarise yourself with the data

Read the transcripts several times. Listen to the recordings again if a passage is unclear. Make notes of first impressions, surprising moments and possible patterns, but do not yet try to force the data into categories. Familiarisation is slow work, and skipping it is the most common cause of shallow analysis. For a thesis with twelve interviews, expect to spend several days on this step alone.

Step 2 — Generate initial codes

A code is a short label that captures one idea in a small piece of data. Coding is done line by line, or sentence by sentence, across the whole data set. A code should stay close to the data: use the participant's own words where possible, and keep codes specific rather than vague. Good codes: cannot afford transport; feels judged by other students; teacher explains slowly. Poor codes: problems; feelings; teaching. The aim is to code everything that might be relevant to the research question, not only the parts that confirm what the researcher already believed.

Step 3 — Search for themes

Once the whole data set is coded, group codes that belong together into candidate themes. A theme is a broader pattern that captures something important about the data in relation to the research question. Several codes — cannot afford transport; evening bus stops early; misses last class — might form a theme called Practical barriers to attendance. Codes that do not fit anywhere go into a miscellaneous group for now; do not discard them yet, because they may form a theme later or challenge an emerging pattern.

Step 4 — Review the themes

Check each theme against the coded extracts and against the whole data set. Does the theme hold together? Is there enough evidence for it? Does it overlap too much with another theme? This step usually splits one broad theme into two, merges two thin ones, or drops a theme that has only one or two weak examples. A theme should be supported by several participants and several extracts, not by a single vivid

quotation.

Step 5 — Define and name the themes

Write a short definition for each theme: what it is, what it is not, and what story it tells. Give it a clear, concise name that captures its essence — Practical barriers to attendance is better than Barriers, and Learning alone at night is better than Night. The definition is what keeps the analysis honest: if a new extract does not fit the definition, it belongs elsewhere.

Step 6 — Write up the analysis

The write-up presents each theme in turn, with a description, an interpretation, and well-chosen quotations that show the evidence. Quotations should be presented with a participant identifier (Participant 4, teacher, aged 28), not with a real name, and they should be selected because they illustrate the theme, not because they are dramatic. The write-up should also discuss what was surprising, what did not fit, and how the themes answer the research question.

Worked example: coding three short extracts

Imagine a study of why adult learners leave evening classes. Three participants say:

- Participant 2: The last bus leaves at nine, and my class ends at half past nine. I have to walk home, and in winter it is dark and cold.
- Participant 5: The fees are not much, but the books cost more than I expected, and then there is the bus fare every night.
- Participant 7: My teacher is kind, but when I ask a question, the others look at me. I feel like I am slowing everyone down.

Initial codes might be: bus leaves before class ends; walking home in dark; books cost more than expected; bus fare every night; feels judged by classmates; fears slowing others down. Grouping them gives two candidate themes. The first four codes form a theme, The hidden costs of attending — transport and money — while the last two form a second theme, Feeling out of place in the classroom. A careful analyst would then check the full data set: if many participants mention money and transport, the first theme is strong; if only one mentions feeling judged, the second theme needs more evidence or must be reported as a smaller, tentative finding. That checking — not the first grouping — is what makes thematic analysis rigorous.

Quality check for thematic analysis

Ask yourself: Have I coded the whole data set, not only the interesting parts? Is each theme supported by more than one participant? Have I reported what did not fit, as well as what did? Could another researcher follow my codes and themes and see how I got there? If the answer to any of these is no, the analysis is not yet finished.

Chapter 5: Quantitative Research: Surveys, Experiments and Correlational Studies

Quantitative research collects numerical data and analyses it with statistics. It is the right choice when a question asks how much, how many, how often, or whether one thing is related to another. Its strength is that it can describe patterns across many people, test a hypothesis, and — in the case of experiments — support a claim about cause and effect. Its weakness is that it can show a pattern without explaining why the pattern exists.

Surveys

A survey collects the same information from many people, usually through a questionnaire. Surveys can describe a population (what proportion of students use the library?), compare groups (do evening students report more stress than day students?), and test relationships (is stress related to attendance?). A good questionnaire uses clear, simple language, one idea per question, and a mix of closed questions (with fixed answers, such as a scale from strongly agree to strongly disagree) and a few open questions. It must be piloted — tried out on a small group first — because a question that seems clear to the researcher is often misunderstood by respondents. Common faults are double-barrelled questions (Do you find the teaching good and the timetable convenient?), leading questions (Do you agree that our excellent library helps you?), and questions that assume knowledge the respondent does not have.

Experiments

An experiment tests whether changing one thing causes a change in another. The researcher manipulates an independent variable (for example, a new teaching method), measures a dependent variable (for example, test scores), and controls other influences as far as possible. The strongest design is the randomised controlled trial: participants are assigned by chance to an experimental group, which receives the new method, and a control group, which receives the usual method, and the two groups are then compared. Random assignment makes the groups similar at the start, so any later difference is more likely to be due to the intervention than to a pre-existing difference.

In education and social research, true experiments are often impractical or unethical — a researcher cannot randomly deny some students a service they need. In that case a quasi-experiment is used: the groups already exist (two classes, two clinics), and the researcher compares them, while acknowledging that the groups may have differed from the start. The methodology chapter must be honest about that limitation, and must not claim a causal effect that the design cannot support.

Correlational studies

A correlational study measures two or more variables and asks whether they vary together. For example, does the number of hours students study correlate with their examination scores? A positive correlation means that as one rises, the other tends to rise; a negative correlation means that as one rises, the other tends to fall. Correlation is useful for prediction and for testing theories, but it does not by itself prove causation. Hours of study and examination scores may both be caused by motivation, or by prior ability. The phrase correlation does not imply causation is a warning every thesis writer should remember, and a correlational study should never be written up as if it had proved a cause.

Variables, hypotheses and measurement

Quantitative research depends on turning ideas into measurable variables. An independent variable is the one the researcher changes or groups by; a dependent variable is the outcome measured. A hypothesis is a specific, testable prediction: Students who attend the revision sessions will score higher on the final test than students who do not. Good hypotheses are clear, testable and falsifiable — it must be possible, in principle, for the data to show they are wrong. Measurement matters too: a questionnaire scale or a test must measure what it claims to measure (validity) and do so consistently (reliability), both of which are discussed in Chapter 9.

Chapter 6: Quantitative Data Analysis: Descriptive and Inferential Statistics in Plain Words

Statistics frightens many thesis students, but the ideas behind it are straightforward. Descriptive statistics describe the data you collected. Inferential statistics use that data to make a careful judgement about a larger population, or about whether a pattern is likely to be real rather than a chance result. You do not need to derive formulas by hand; you need to choose the right test, run it correctly, and interpret the result honestly.

Descriptive statistics

Start every quantitative analysis by describing the data. For a numerical variable, report the mean (the average), the median (the middle value when the data are ordered), and the mode (the most common value). The mean is familiar but can be misleading when a few very high or low values pull it away from the typical case; the median is often a better guide when data are skewed, as income data usually are. Also report the spread: the range (lowest to highest) and the standard deviation, which tells you, roughly, how far a typical value sits from the mean. A small standard deviation means the values cluster together; a large one means they are spread out.

Inferential statistics in plain words

Inferential statistics ask: if I see a difference or a relationship in my sample, how likely is it to be true in the population, and how likely is it to have happened by chance? The two ideas that matter most are the p-value and the confidence interval. A p-value is the probability of seeing a result at least as large as the one observed, if in reality there is no effect at all. By convention, a p-value below 0.05 is called statistically significant, meaning the result is unlikely enough to be worth taking seriously. But statistical significance is not the same as importance: with a very large sample, a tiny and meaningless difference can be statistically significant, while with a small sample a large and important difference may not reach significance. Always report the size of the effect as well — how many marks higher, how many percentage points different — so the reader can judge its practical meaning.

Choosing a test — a simple guide

Worked example — reading a result

A thesis reports: Students who attended revision sessions scored a mean of 68 (standard deviation 9), while those who did not scored a mean of 61 (standard deviation 10). The difference of 7 marks was statistically significant (p = 0.01, 95% confidence interval 2 to 12 marks). A good write-up says: the difference is unlikely to be due to chance, the effect is about 7 marks, and the true effect could plausibly be as small as 2 or as large as 12 marks. A poor write-up says only: the result was significant. The first tells the reader what the finding means; the second hides it.

Chapter 7: Mixed Methods Research: Designs and When to Use Them

Mixed methods research combines qualitative and quantitative data in a single study. It is not simply doing two separate studies side by side; the two kinds of data must be brought together so that each one helps explain or strengthen the other. The approach is growing rapidly in thesis research because many real questions have both a how many part and a why part, and neither numbers nor interviews alone can answer both.

The three main designs

In an exploratory sequential design, the order is reversed: qualitative work comes first, to explore a little-known topic, and the findings are then used to build a quantitative instrument, such as a questionnaire. Interviews might reveal the reasons students give for leaving, and those reasons then become the items in a survey given to a larger group, so the researcher can find out how widespread each reason is. This design is ideal when no existing questionnaire fits the topic, but it is the most demanding, because it requires both qualitative analysis and questionnaire design.

When to use mixed methods — and when not to

Use mixed methods when your question genuinely has two parts that need different kinds of evidence, when one kind of data alone would leave an important question unanswered, or when you need to check a finding from one method against another. Do not use mixed methods simply to make a thesis look more impressive, or because a supervisor suggested adding a few interviews to a survey. A mixed methods thesis is more work than a single-method thesis, it needs skill in two traditions, and examiners will judge both parts. A well-done single-method study is always better than a poorly integrated mixed methods study.

Practical tip

If you choose mixed methods, draw a simple diagram of your design — two boxes for the two phases, with arrows showing the order and the point where the findings are integrated. Put the diagram in your methodology chapter. Examiners find it very helpful, and it forces you to be clear about your own plan.

Chapter 8: Sampling: Who to Study and How Many

Sampling is the process of choosing whom to study. No thesis can study everyone, so every study studies a sample, and the quality of the sample decides how far the findings can be trusted. Two questions matter: who is included, and how many are included. The answers depend on the approach, the question and the resources available.

Probability and non-probability sampling

In probability sampling, every member of the population has a known chance of being selected, usually through random selection. This is the basis for generalising from a sample to a population in quantitative research. In non-probability sampling, selection is not random, and the aim is usually not statistical generalisation but depth, relevance or access. Both are legitimate; the mistake is to use one and claim the benefits of the other. A convenience sample of volunteers cannot support a claim about a whole population, however large it is.

How many is enough?

There is no single magic number. For quantitative research, sample size depends on the size of the effect you hope to detect, the variability in the population, and the precision you need. Larger samples give more precise estimates and more power to detect real effects, but they also cost more time and money. A power calculation, done before data collection with the help of a statistics adviser or a standard formula, gives a defensible target number, and examiners expect to see it or a clear justification in its place. For qualitative research, the guiding idea is saturation: keep interviewing until new interviews add little new information and the same themes

recur. In practice, many interview studies reach saturation between twelve and twenty participants, but the number depends on the diversity of the sample and the breadth of the question. A study of a very narrow, homogeneous group may saturate sooner; a study comparing several different groups will need more.

Table 1: Comparison of common sampling methods

Method	Type	Best for	Main risk
Random	Probability	Generalising to a population	Needs a complete list
Stratified	Probability	Representing subgroups fairly	Needs group information
Cluster	Probability	Studying whole groups, e.g. schools	Groups may differ greatly
Purposive	Non-probability	Depth; expert or typical cases	Cannot generalise statistically
Convenience	Non-probability	Quick, low-cost access	Bias; unrepresentative sample
Snowball	Non-probability	Hard-to-reach groups	Similar people refer each other

Remember

A large biased sample is worse than a small careful one. A thousand volunteers from a single social media post tell you less about a population than a hundred people chosen at random from it.

Chapter 9: Validity, Reliability and Trustworthiness

Readers of a thesis ask a simple question: can I trust these findings? Quantitative and qualitative researchers answer that question with different words, but the concern is the same. Quantitative research speaks of validity and reliability. Qualitative research speaks of trustworthiness, with the criteria of credibility, transferability, dependability and confirmability. A good thesis explains, in its own terms, what was done to earn the reader's trust.

Validity and reliability (quantitative)

Validity asks whether a study measures what it claims to measure. A test of reading ability that mainly tests general knowledge is not a valid measure of reading. Internal validity asks whether the study's own conclusions are sound: did the intervention really cause the change, or could something else explain it? External validity asks whether the findings apply beyond the study's own sample and setting. Reliability

asks whether the measurement is consistent: would the same instrument give similar results if used again, or by a different researcher? A bathroom scale that gives a different reading every time you step on it is unreliable, however plausible each single reading looks. In a thesis, reliability is often shown by reporting a reliability coefficient for a questionnaire scale, or by describing how an instrument was piloted and refined.

Trustworthiness (qualitative)

Credibility is the qualitative equivalent of internal validity: are the findings a believable account of the participants' experience? It is strengthened by prolonged engagement with the setting, by triangulation — using more than one method, source or researcher to check a finding — and by member checking, returning transcripts or summaries to participants and asking whether they recognise themselves in them. Transferability is the equivalent of external validity: could the findings apply elsewhere? It is supported by thick description — a rich, detailed account of the setting and participants — so that readers can judge for themselves whether the findings transfer to their own context.

Dependability asks whether the research process was consistent and well documented, so that another researcher could follow the trail. An audit trail — dated notes, coding decisions, drafts of themes, and a record of changes of mind — is the practical way to show dependability. Confirmability asks whether the findings are grounded in the data rather than in the researcher's preferences. Reflexivity helps here: a short account of the researcher's own background, assumptions and relationship to the participants, and how those might have shaped the study. A researcher who taught the students they interviewed, for example, should say so and explain how they handled the power difference.

Chapter 10: Ethics in Research

Ethics is not a form to be filled in after the design is finished. It is part of the design itself, and it should be considered from the first planning meeting. Research ethics protects participants, protects the researcher, and protects the credibility of the findings. Most universities require ethical approval before data collection begins, and collecting data without it can mean the data cannot be used in the thesis at all. Always check your institution's rules first, and allow time for approval — it often takes weeks.

Informed consent

Participants must understand what the study is about, what they will be asked to do, how their data will be used, and that they are free to refuse or to withdraw at any time without giving a reason and without penalty. Consent should be given freely, on the basis of clear information, and recorded — usually in writing, or by audio

recording for participants who cannot read the form. For children and other vulnerable groups, consent must come from a parent or guardian as well as assent from the participant themselves. Deception — misleading participants about the purpose of a study — is only rarely justifiable, and only with strong reasons and a full debriefing afterwards.

Confidentiality and anonymity

Confidentiality means that information given in confidence will not be shared in a way that identifies the person. Anonymity means that even the researcher cannot link a response to a named person, as in an unnamed questionnaire. Interviews are rarely anonymous, because the researcher knows who spoke, but they must be confidential: names are replaced with codes or pseudonyms, identifying details are removed from quotations, and recordings and transcripts are stored securely, with access limited to the research team. Data protection rules in your country govern how personal data are stored and for how long, and the methodology chapter should state what was done to comply with them.

Harm, benefit and honesty

Research must not harm participants. Harm includes physical harm, but also embarrassment, distress, loss of reputation, or loss of time and income. A study that asks people to recall traumatic experiences must have support available and must allow participants to stop at any point. The researcher must also weigh benefit: the study should be worth doing, and its findings should be shared with those who took part, where appropriate, in a form they can use. Finally, research must be honest: data must not be fabricated, results must not be selectively reported to favour a preferred conclusion, and the work of others must be acknowledged. Plagiarism, including presenting another's ideas as your own, is an ethical failure as well as an academic one.

Ethics checklist — before you collect any data

Have you obtained ethical approval? Is your consent form clear and in plain language? Is your data storage secure? Have you planned how to anonymise quotations? Have you thought about what you will do if a participant becomes distressed? If the answer to any of these is no, do not start collecting data yet.

Chapter 11: Choosing the Right Approach for Your Question

The decision that matters most is the one made earliest: which approach fits the research question? The guide below is a practical decision aid, not a set of rigid rules, but it captures the reasoning examiners expect to see. Work through it with your own question in mind, and write down your answers — they will form the core of your

methodology chapter's justification.

Step 1 — What kind of question is it?

Sort your question into one of four types. A what question that asks how widespread something is (What proportion of students use the library after 6 p.m.?) points towards a survey. A whether or how much question that asks about a relationship or a difference (Does extra tuition improve scores? Is stress related to attendance?) points towards quantitative work, and towards an experiment if a causal claim is wanted. A how or why question that asks about meaning, experience or process (How do students decide to leave? Why do teachers resist a new curriculum?) points towards qualitative work. A question with two parts — how many, and why (How many leave, and what reasons explain the pattern?) — points towards mixed methods.

Step 2 — What is already known?

Step 3 — What is feasible?

Step 4 — Write the justification

Write a paragraph that follows this pattern: My question asks [type of question]. Little is known about [topic] in this setting, so [approach] is appropriate because [reason]. The alternatives were considered: [alternative] would not answer the question because [reason]. This paragraph, expanded with a few sentences about the paradigm and the specific design, is the heart of a methodology chapter. If you cannot write it, your design is not yet clear enough.

Worked example — three questions, three decisions

Question A: What proportion of final-year students have used AI tools in their assignments? Decision: survey — the question asks how widespread a behaviour is. Question B: How do students decide when using an AI tool is acceptable? Decision: interviews — the question asks about judgement and meaning. Question C: How many students use AI tools, and how do they judge when it is acceptable? Decision: mixed methods — survey first, interviews second to explain the pattern.

Chapter 12: Common Mistakes and a Final Checklist

The most common mistakes

1. Mismatched question and method. A question that asks why is answered with a survey that can only show how many, or a question that asks how many is answered with five interviews. The fix is to check the match before collecting data, using Chapter 11.
2. Describing tools instead of justifying choices. Pages of detail about a

questionnaire, with no explanation of why a questionnaire was the right tool. The fix is to give a reason for every major choice, and to name the alternatives that were considered and rejected.

3. No pilot study. A questionnaire or interview guide used for the first time on the real sample, with ambiguous questions discovered too late. The fix is to pilot with a small group and report what changed as a result.
4. Sampling that cannot support the claim. A convenience sample used to claim results for a whole population, or a single case used to claim a universal rule. The fix is to claim only what the sample allows, and to say so honestly.
5. Over-claiming from small samples. A difference of a few marks in a sample of twenty presented as proof of a large effect. The fix is to report effect sizes and confidence intervals, and to discuss uncertainty.
6. Ignoring what did not fit. Quotations or results that contradict the main finding left out, so the analysis looks tidier than the data were. The fix is to report negative cases and discuss them.
7. Ethics as an afterthought. Data collected before approval, consent forms missing, or identifying details left in quotations. The fix is to get approval first and to check anonymity before submission.
8. Analysis that does not match the data. Statistical tests used on data that break their assumptions, or themes announced with no evidence from the transcripts. The fix is to check assumptions, and to show the evidence for every theme.
9. Copying a previous thesis. A methodology copied from another study, with details that do not fit the current question, setting or sample. Examiners recognise this at once. The fix is to write your own account, from your own decisions.
10. No limitations section. A thesis that claims its study has no weaknesses is not believed. The fix is to discuss limitations openly and to suggest what future research should do differently.

Final checklist — use before submission

- Research question restated at the start of the methodology chapter.
- Paradigm named, in one short paragraph, and linked to the question.
- Design named and justified; alternatives considered and reasons given.
- Participants and sampling described: who, how many, how chosen, response rate.
- Instruments described and piloted; changes after piloting reported.
- Data collection procedure described step by step, in enough detail to repeat.
- Analysis procedure described step by step, with the test or analytic method named.

- Validity, reliability or trustworthiness measures described.
- Ethical approval, consent, confidentiality and data storage described.
- Limitations discussed honestly, with implications for the findings.
- Every claim in the findings can be traced back to data shown in the thesis.
- All sources cited; no claim presented as your own that belongs to someone else.

Chapter 13: Conclusion

Methodology is not a hurdle to be cleared on the way to the findings. It is the part of a thesis that earns the findings their authority. A clear question, a paradigm that fits it, a design chosen for a stated reason, careful sampling, honest analysis and ethical care — these are what separate research from opinion, and a thesis from an essay. The details differ between qualitative, quantitative and mixed methods research, but the standard is the same in all three: show your work, justify your choices, and be honest about what your evidence can and cannot show.

For the student writing a thesis, the practical advice of this guide can be summarised in a few sentences. Start from the question, not from a favourite method. Choose the approach that answers that question, and be able to explain why in a paragraph. Pilot your instruments. Sample carefully and report the sample honestly. Analyse the whole data set, including what does not fit. Report uncertainty as well as results. Discuss your limitations. And keep an audit trail — notes, decisions, drafts — so that you can answer any question an examiner asks about how the study was done.

Research is a craft, and like any craft it improves with practice and with honest feedback. The first thesis is rarely perfect, and examiners do not expect perfection. They expect clarity, rigour and honesty, and a methodology chapter that shows the researcher understood what they were doing and why. If this guide helps a student write that kind of chapter, it has done its job.

Suggested Reading

The books below are well-known, widely used introductions to research methodology. They are listed as general guidance for further reading; a student should consult the edition held by their own university library and follow their department's referencing style for full details.

- Creswell and Creswell — Research Design: Qualitative, Quantitative, and Mixed Methods Approaches. A clear introduction to the three approaches and to designing a study, widely used in thesis courses.
- Bryman — Social Research Methods. A comprehensive and readable overview of research design, sampling, data collection and analysis in the social sciences.
- Saunders, Lewis and Thornhill — Research Methods for Business Students. A practical guide to choosing a research philosophy, design and methods, with worked examples from business and management.
- Braun and Clarke — Thematic Analysis: A Practical Guide. The standard practical introduction to thematic analysis, with step-by-step guidance and examples.
- Field — Discovering Statistics Using IBM SPSS Statistics. A friendly, thorough introduction to quantitative analysis and statistics for students who do not come

from a mathematics background.

- Yin — Case Study Research: Design and Methods. The classic guide to designing and carrying out case studies, including how to handle evidence and generalisation.
- Silverman — Doing Qualitative Research. A practical and thoughtful guide to qualitative design, interviewing, observation and analysis.
- Robson and McCartan — Real World Research. A strong guide to applied research design, ethics, sampling and evaluation, especially useful for professional and practice-based theses.

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